

AE 8883: Spacecraft Engineering I

Electrical Schematics

September 26, 2022

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Outline of today's seminar

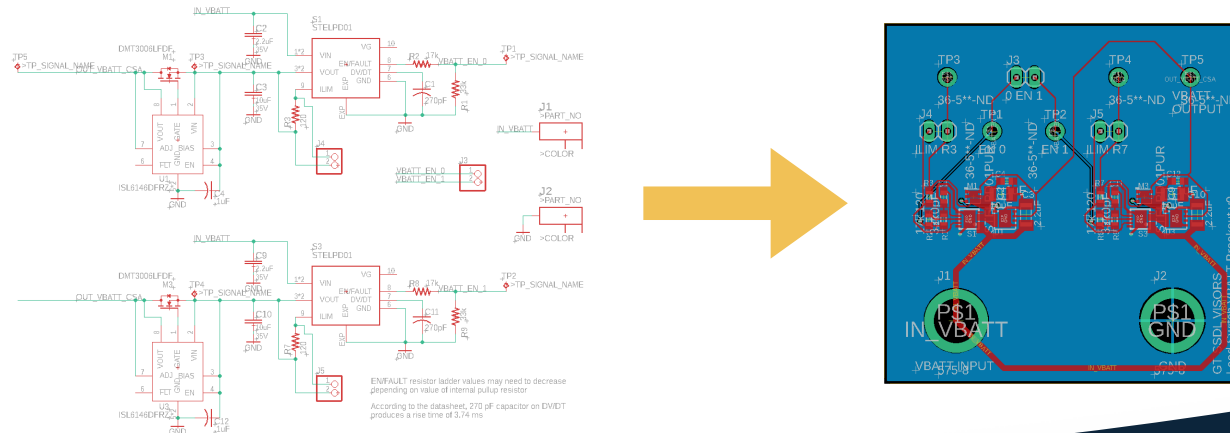
- ⦿ EDA Software and Electrical Schematics
- ⦿ Electrical Schematic Elements
 - Symbols
 - Nets
 - Names and Labels
 - Title Blocks
 - Comment Blocks
- ⦿ EDA Software Tools
 - ERC
 - Netlist Generation
 - BOM Export

EDA/ECAD Software

Software focused on designing electronic circuits and PCBs (printed circuit boards)

- EDA = Electronic Design Automation
- ECAD = Electronic Computer-Aided Design

General workflow involves the creation of an electrical schematic, followed by a layout of all circuit components



Electrical Schematics

2D logical representation of an electric circuit

- ⦿ What passive devices and ICs (integrated circuits) are present in this circuit, and in what quantities?
- ⦿ What are the values & specifications of these passives & ICs?
- ⦿ What connections are present? Which pins are connected to which other pins?

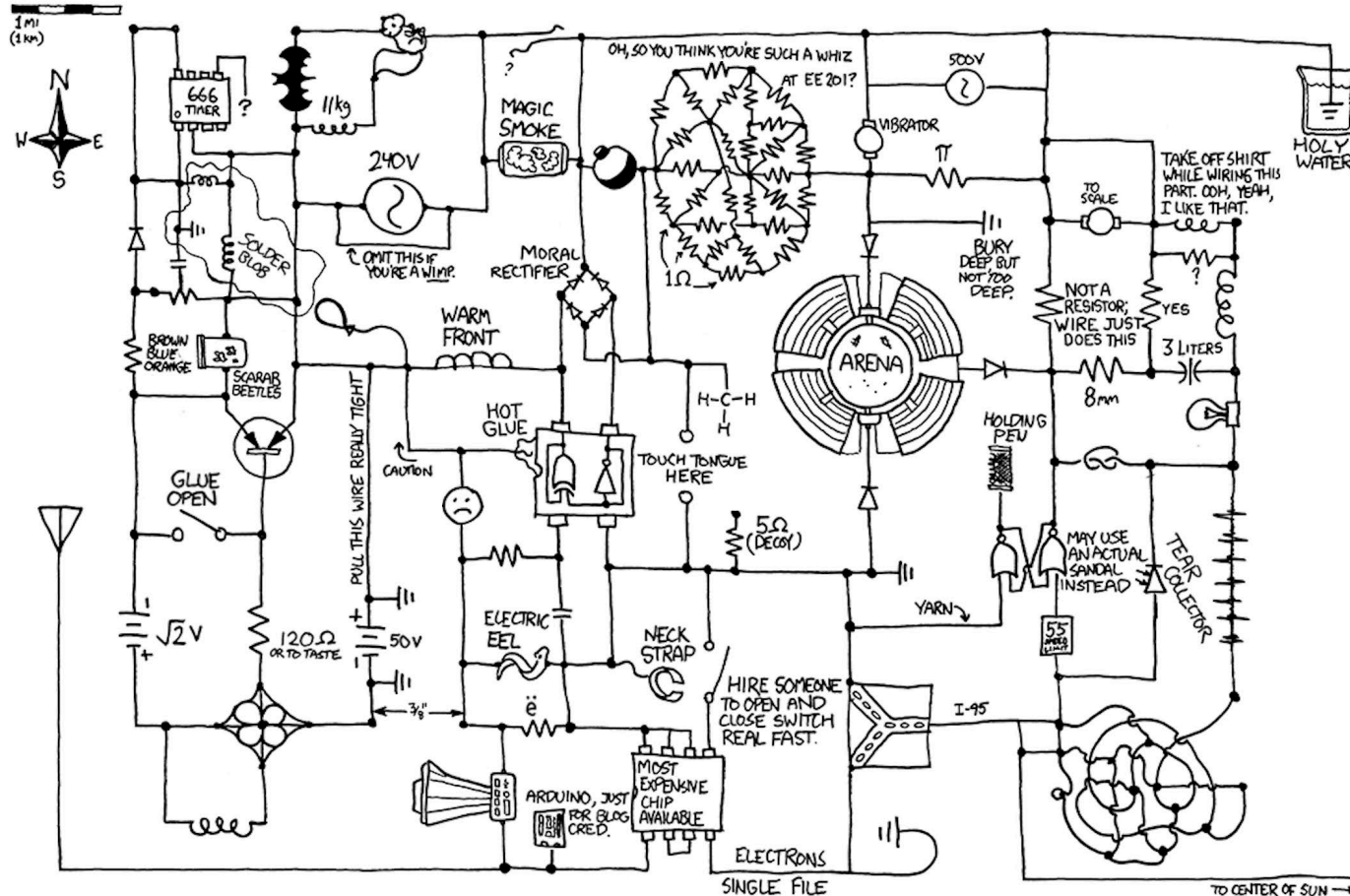
Note: **electrical schematics** differ from **wiring diagrams**

- ⦿ Wiring Diagram
 - How are passives and ICs physically laid out in the circuit?
 - What wires are run between various components in the circuit?
 - What are the lengths of each wire run?



Working with Electrical Schematics

Electrical Schematic Elements

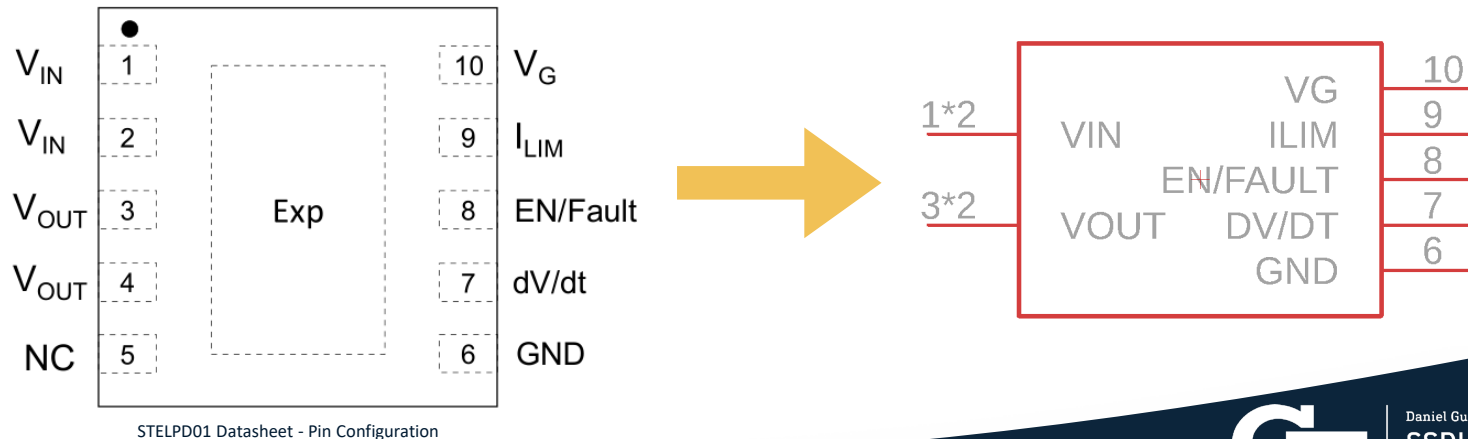


Note: Not an electrical schematic

<https://redd.it/62wz02/> based on <https://xkcd.com/730/>

Electrical Schematic Elements - Symbols

- Abstract representations of passive components and ICs
 - Any component you'll need to solder to the board should generally be represented by a symbol on the electrical schematic
 - Displays connections that can be made to it as **pins**
- IC schematics can be laid out however is intuitive
- Commonly used devices have generic symbols they're associated with, based on the standard you're following



Electrical Schematic Elements - Symbols

Symbols for the same component may vary between different standards

- IEEE/ANSI 315-1975

- <https://standards.ieee.org/ieee/315/515/>

- (Institute of Electrical and Electronics Engineers)
 - (American National Standards Institute)
 - This standard was inactivated in 2019, but is still widely used in the U.S.

- IEC 60617

- <https://webstore.iec.ch/publication/2723>

- (International Electrotechnical Commission)

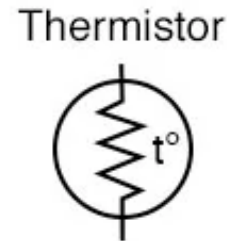
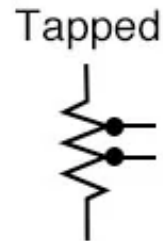
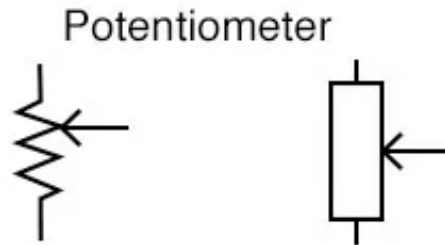
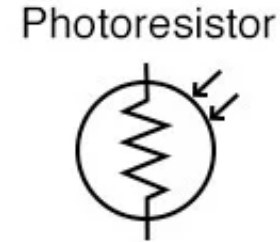
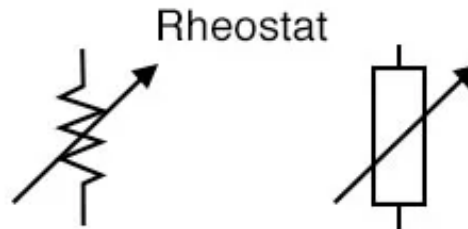
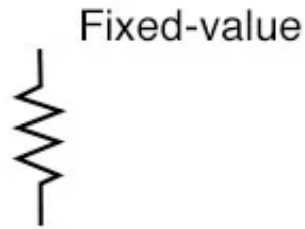
- NEMA ICS 19-2002

- <https://www.nema.org/standards/view/diagrams-device-designations-and-symbols>

- (National Electrical Manufacturers Association)

Electrical Schematic Elements - Symbols

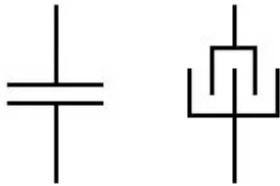
Resistors



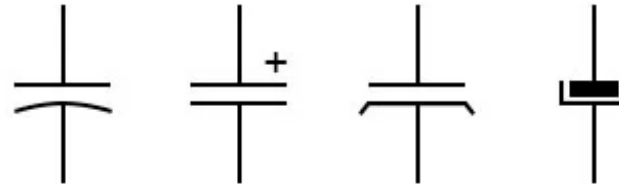
Electrical Schematic Elements - Symbols

Capacitors

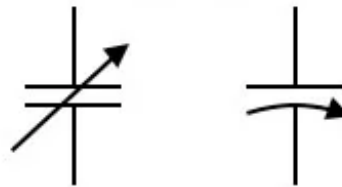
Non-polarized



Polarized (top positive)

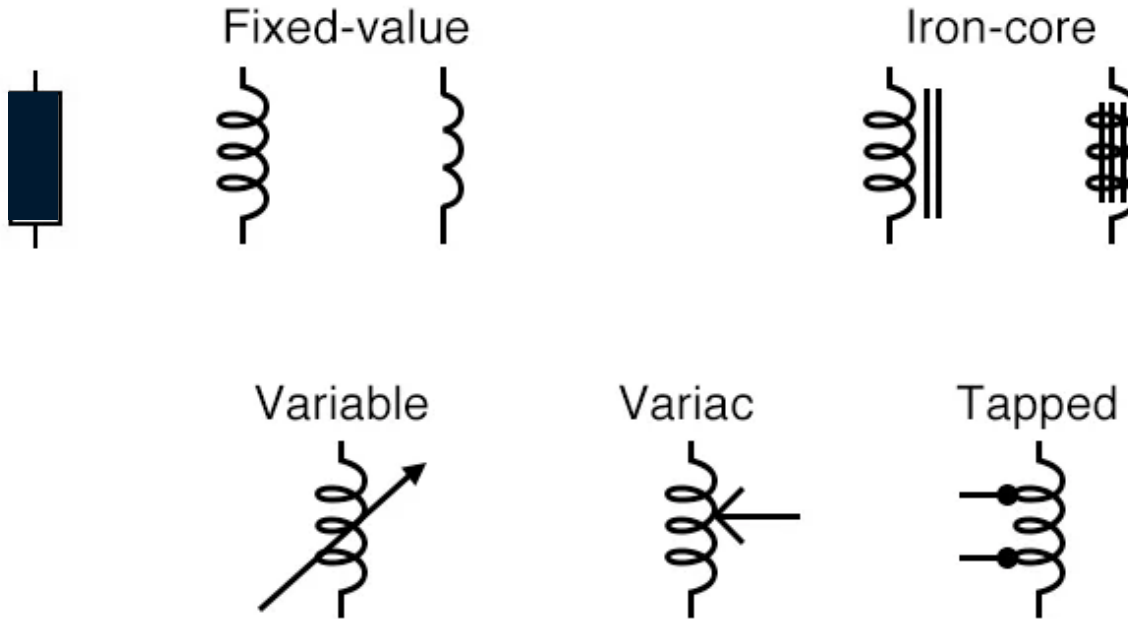


Variable



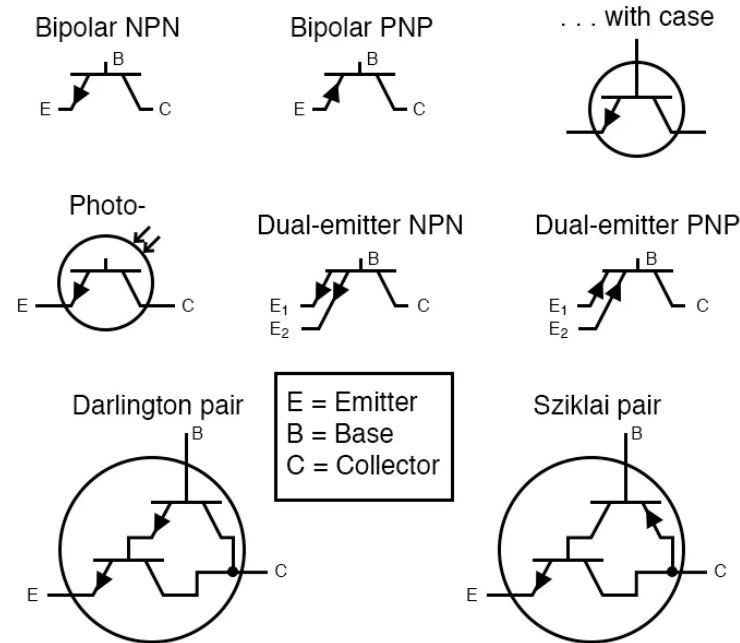
Electrical Schematic Elements - Symbols

Inductors

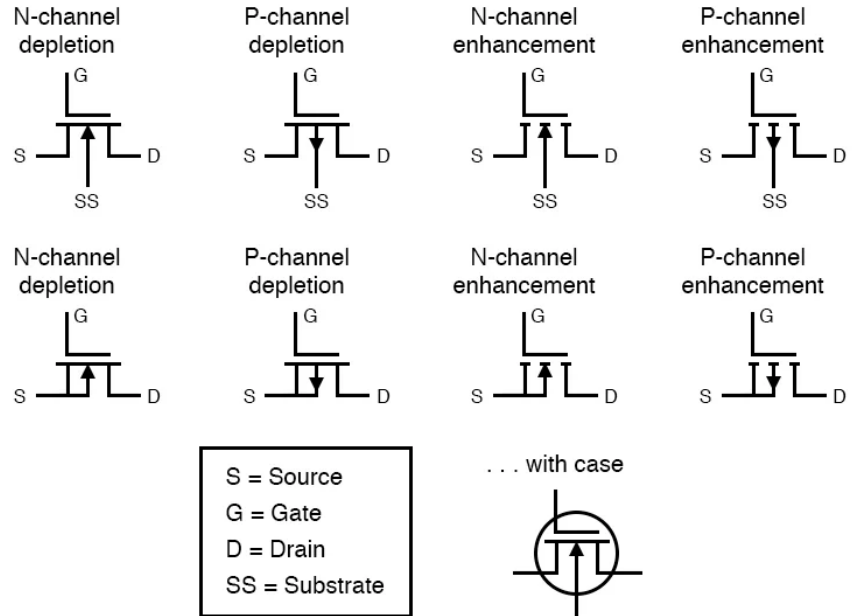


Electrical Schematic Elements - Symbols

BJTs (Bipolar Junction Transistors)



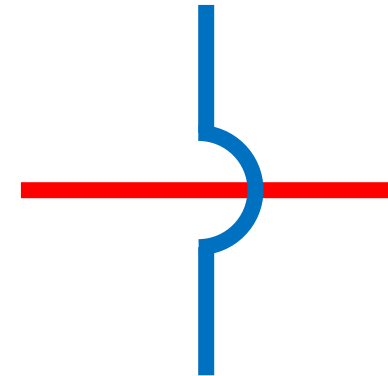
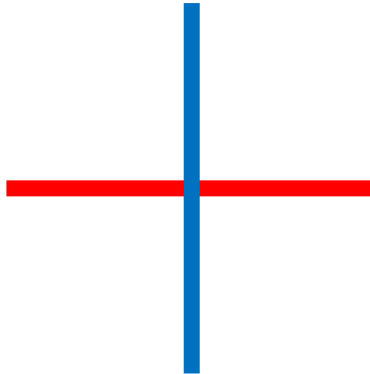
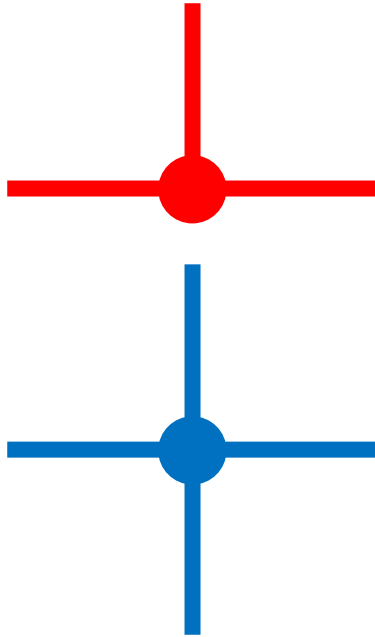
MOSFETs (Metal-Oxide-Semiconductor Field-Effect Transistors)



Electrical Schematic Elements - Nets

Represent electrical connections between pins

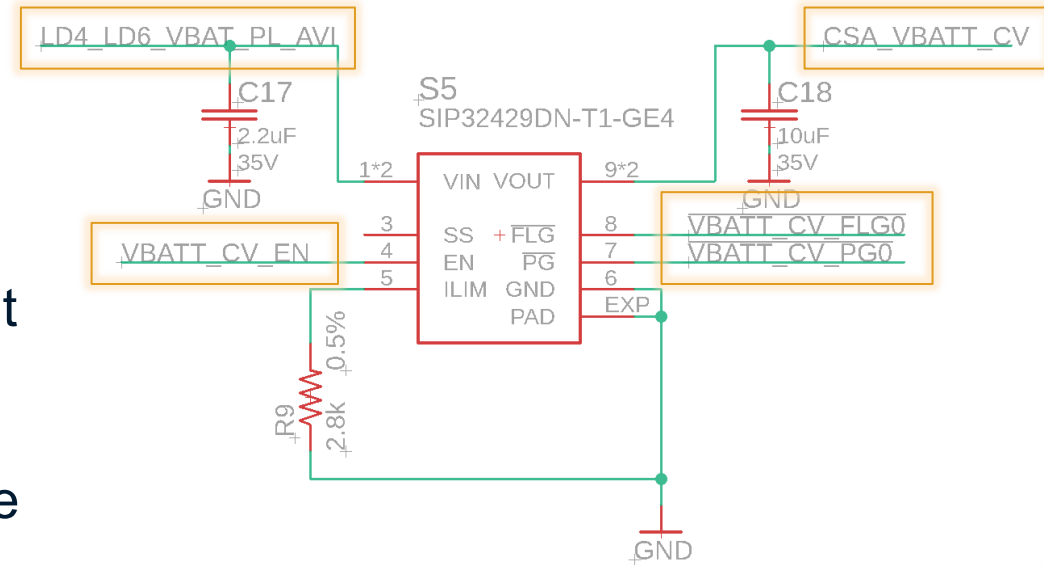
- ⦿ A dot at an intersection signifies the nets are joined (if there's no dot, the nets belong to separate signals)



(older convention, avoid)

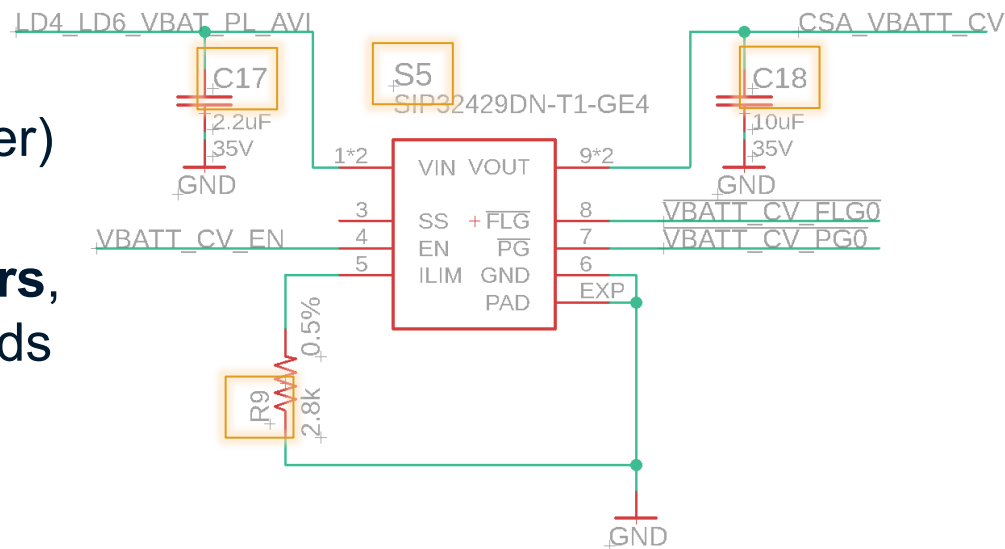
Electrical Schematic Elements - Names & Labels

- Nets can be assigned names for clarity and organization
- **On the same sheet, different nets with the same name are electrically connected**
 - Done to avoid rat nests of nets cluttering up the schematic
 - If nets have the same name, but are on different sheets, whether they are connected will vary depending on your choice of EDA software



Electrical Schematic Elements - Names & Labels

- Every passive component and IC on a schematic is assigned its own unique name
 - Generally consists of a letter prefix and number
 - Aids assembly process & BOM (Bill of Materials) generation (more on that later)
 - Prefixes used for names are called **reference designators**, and they also follow standards (kinda... just be consistent)



Electrical Schematic Elements - Names & Labels

- ◎ IEEE 200-1975 (ANSI Y32.16-1975)
<https://ieeexplore.ieee.org/document/4483758>
- ◎ ASME Y14.44-2008
<https://www.asme.org/codes-standards/find-codes-standards/y14-44-reference-designations-electrical-electronics-parts-equipment/2008/drm-enabled-pdf>
 - (American Society of Mechanical Engineers)
 - Designed to replace IEEE 200-1975

Electrical Schematic Elements - Names & Labels

Common designators (not necessarily standard-compliant)

Designator	Component Type
C	Capacitor
D	Diode
F	Fuse
J	Jack Connector
JP	Jumper
K	Relay/Contactor
L	Inductor/Coil

Designator	Component Type
M	Motor
Q	Transistor
R	Resistor
S	Switch
TP	Test Point
U	IC (Integrated Circuit)
X/Y	Crystal/Oscillator

https://en.wikipedia.org/wiki/Reference_designator#Designators

Again, the important part is to
be consistent within a project



Electrical Schematic Elements - Title Block

Documents information about the current sheet's electrical schematic and the project it belongs to

- Project Name
- Description of Schematic Circuits on Sheet
- Designer Name
- Date
- Sheet Number out of Total (if part of a multi-sheet schematic)
- Design Revision Number

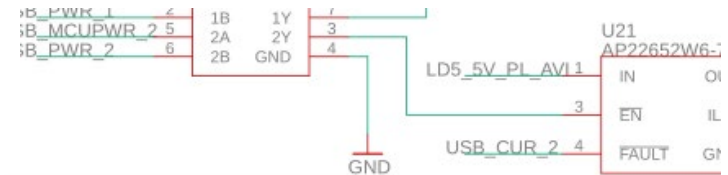
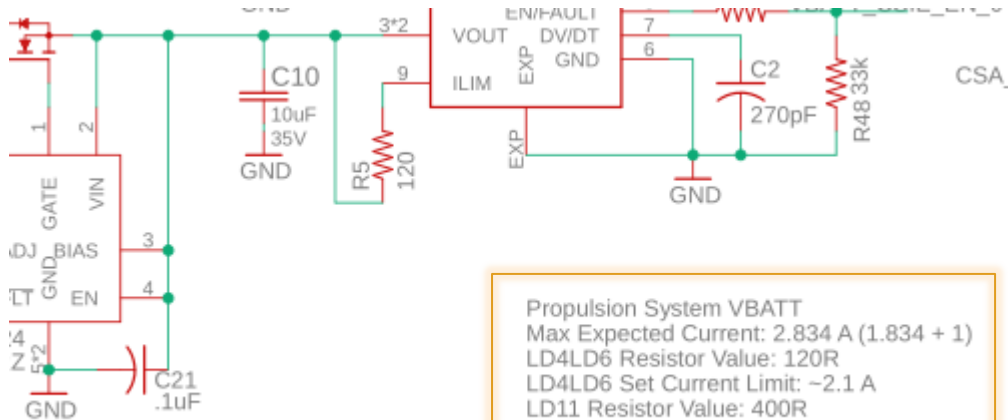
Project: GT1		Main Board: Onion Omega2S		D
Georgia Tech Space Systems Design Laboratory		Designer: Andrew Loomis		
		1/17/20		
		Sheet: 3/11	Rev: 1.0	
4	5	6		



Electrical Schematic Elements - Comment Blocks

Useful for giving additional context to design decisions

- Many circuit design decisions may be based to some degree off of specifications listed in datasheets and documentation for that IC
- Make notes of important decisions in these text blocks
- Useful during design reviews!** (don't make your reviewers dig through datasheets during your review)



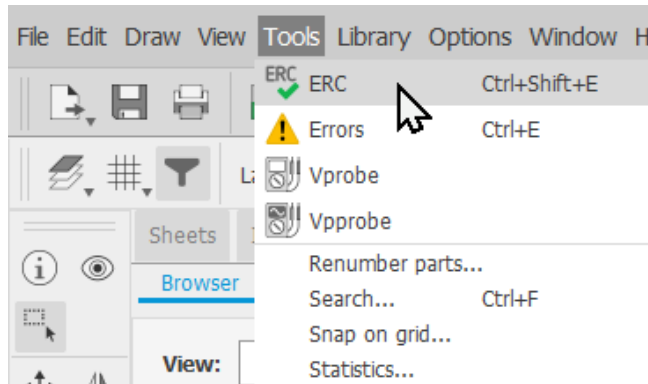
Note that this is technically a non-compliant USB implementation. Maybe OK because ports 1,2 have permanently wired devices w/ known current draw?



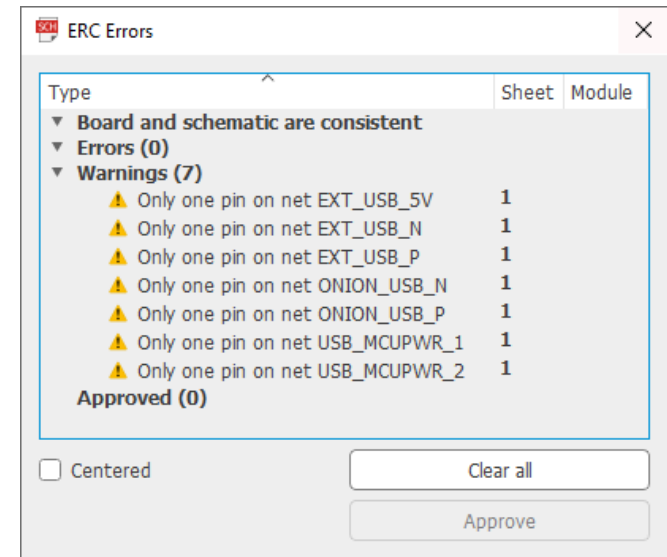
EDA Software Tools

ERC (Electrical Rules Check)

- Upon completion of your electrical schematic, you'll want to have your software help you validate your circuit by checking for errors and violations
- You should have no Errors
- Evaluate Warnings on a case-by-case basis



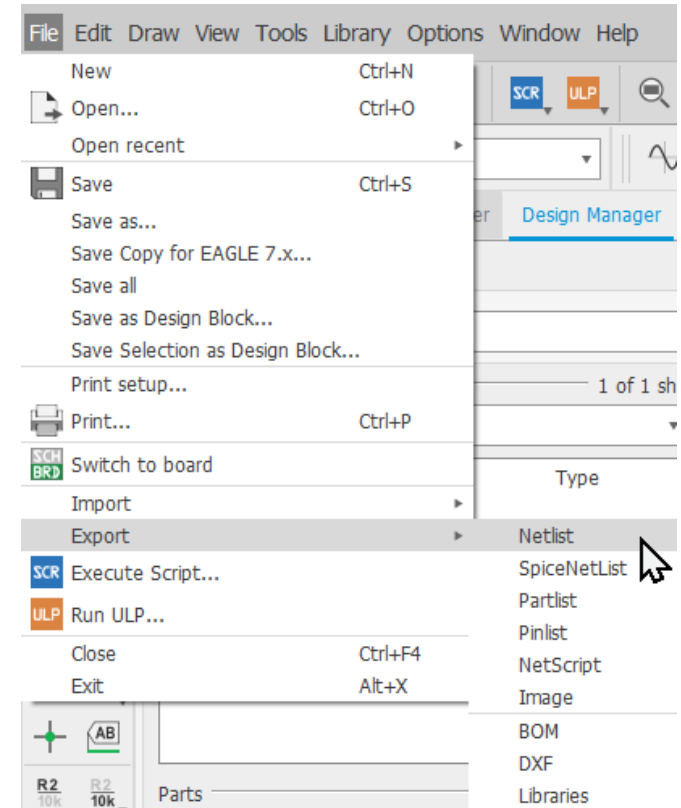
ERC in EAGLE



ERC Error window in EAGLE

Netlist Generation

- The nets in your schematic can be compiled into formatted text files containing nets and net labels, reference designators, and pin numbers
- Useful if you want to run tests in a circuit simulator (such as SPICE) or export your electrical schematic as a generic circuit description for different EDA software

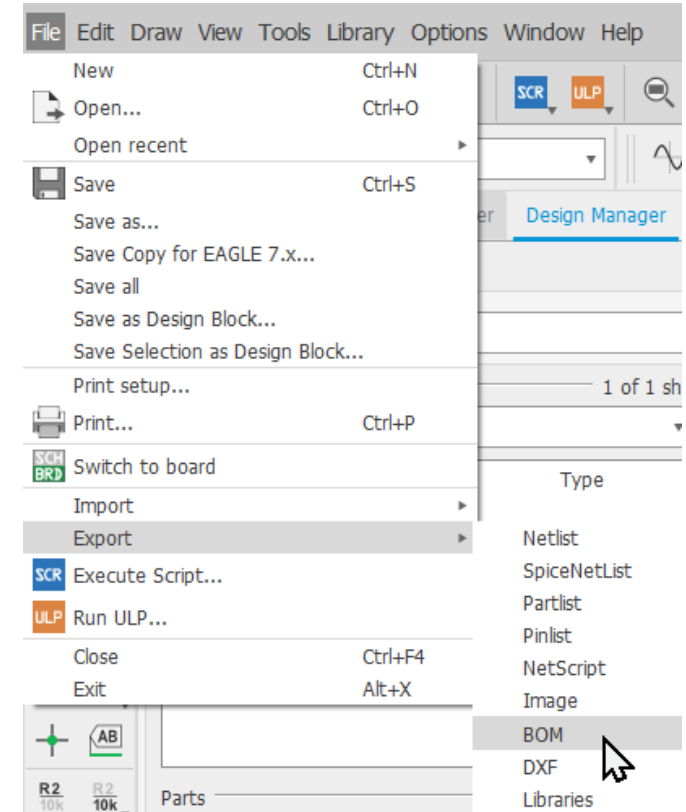


Netlist generation in EAGLE

BOM (Bill of Materials) Export

Comprehensive list of circuit components required to manufacture your circuit

- Either used for your own reference when ordering parts to assemble on your own, or sent to an assembly house so they know what to order
- Components are listed according to their name on the schematic, which makes it easy to figure out their model/value during assembly.



BOM Export in EAGLE

BOM (Bill of Materials) Export

Eagle: Bill Of Material

Current variant: "

Qty	Value	Device	Package	Parts	Description	COLOR	PART_NO
1	ECS-2033-060-BN	ECS-2033-060-BN	ECS-2033	X2			
1	SN74LVC2G32QDCURQ1	SN74LVC2G32QDCURQ1	VSSOP-8	U4	IC GATE OR 2CH 2-INP 8VSSOP		
1	TUSB2036VF	TUSB2036VF	LQFP32-7X7	U\$2			
2	102.3k	R-US_R0603	R0603	R9, R10	RESISTOR, American symbol		
2	575-8	575-8	575-8	J1, J2	Digi-Key Part Number: 36-575-8-ND	RED	36-575-8-
2	SFU3064GC2AE2TO-I-LF-1A1-STD	SFU3064GC2AE2TO-I-LF-1A1-STD	SFU3064GC2AE2TO-I-LF-1A1-STD	U6, U7			
3	AP22652W6-7	AP22652W6-7	SOT-26	U1, U2, U3			
8	22pF	C-USC0603	C0603	C1, C2, C3, C4, C5, C6, C7, C8	CAPACITOR, American symbol		
8	27	R-US_R0603	R0603	R1, R2, R3, R4, R5, R6, R7, R8	RESISTOR, American symbol		

List type: ☐ Parts ☒ Values ☒ List attributes

Output format: ☐ Text ☒ CSV ☐ HTML

View Save... Help Close

Version 1.11

In EAGLE, you can list by Values to get the quantities of each unique part used in your design, useful when you go to order components.

Electrical Schematics Lab:

Van Leer E283

09/26/2022 - 5:00 PM



Autodesk EAGLE Installation

- Login with an Autodesk account with Educational Access, or create one by navigating to the following link and hitting **Get Started**:
<https://www.autodesk.com/education/edu-software/overview>
 - Note: If you aren't able to get Educational Access for your account approved by Monday, don't worry, the features available in the free version of EAGLE will still allow you to do all of the lab
- Download the EAGLE installer for your preferred operating system:
<https://www.autodesk.com/products/eagle/free-download>

Note for Windows 11 users!

- If you are doing a clean install of EAGLE, you will need to make the following modification after installation to get it to open:
 1. Go to your EAGLE installation folder, E.g. **C:\EAGLE9.6.2**
 2. Rename **LIBEAY32.DLL** to **LIBEAY32.DLL.BAK** (make sure the file extension gets modified)